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TRAJECTORY PLANNING OF FIXED-WING UAV USING EXACT PENALTY FUNCTION

Yangzhou Cao and Bin Li*

School of Aeronautics and Astronautics, Sichuan University, Chengdu, China

Abstract. Trajectory planning for fixed-wing unmanned aerial vehicles (UAVs) requires time optimality under complex dynamic and physical constraints, which is critical for task efficiency. To address the high complexity and inaccurate constraint satisfaction of traditional methods in variable time domains and limited states of UAVs, this study proposes an approach integrating control parameterization and exact penalty function. Continuous controls of the commonly used three-degree-of-freedom (3-DOF) dynamic model are converted to finite parameters, and time optimality is considered as the optimization objective. Furthermore, state and control constraints of UAVs are embedded into the objective function via a dynamic penalty parameter. To enable the actual state of the six-degree-of-freedom (6-DOF) UAV model to track the planned trajectory, we impose multiple state and control constraints and conduct replanning within the framework of model predictive control. Experiments on the F-16 aircraft model verify the method's feasibility and effectiveness.

1. Introduction.

1.1. Research background and significance.

Fixed-wing unmanned aerial vehicles (UAVs) [3] have been widely deployed in military reconnaissance, disaster monitoring, geological surveying, traffic inspection, and many other fields, owing to their strong endurance, high flight efficiency, and good payload adaptability. Particularly in complex mission environments such as mountain rescue operations or urban low-altitude penetration, UAVs are required not only to possess autonomous flight capabilities, but also to accomplish predefined tasks within strict time windows and under dynamic constraints. As one of the key technologies enabling autonomous UAV flight, trajectory planning directly determines the efficiency and safety of mission execution.

Time optimality is often a core requirement for UAVs performing urgent tasks. For instance, in battlefield scenarios, a UAV may need to traverse hostile air defense zones in the shortest possible time, or deliver emergency supplies to a target location rapidly. Such missions demand that the system achieve minimal transition time from the initial state to the target state while satisfying multiple physical constraints. However, the dynamics of fixed-wing UAVs are highly complex and

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*Corresponding author: Bin Li.

exhibit strong coupling effects, such as dynamic interactions among velocity, flight path angle, and overload, as well as variations in aerodynamic parameters with flight states, all of which increase the difficulty of trajectory planning.

Furthermore, during actual flight, the UAV must strictly adhere to various state and control constraints, including but not limited to allowable altitude ranges, velocity boundaries, overload capabilities, and control surface effectiveness. Violation of any of these constraints could lead to control failure or even catastrophic incidents. Therefore, achieving time-optimal trajectory planning while simultaneously satisfying multiple constraints within a highly coupled nonlinear dynamic system has become a challenging and active research topic in the field of autonomous UAV control. Research in this area not only carries significant theoretical value, but also considerable engineering importance for enhancing the mission performance of UAVs in complex environments.

1.2. **Research status and limitations.** Currently, trajectory planning methods for fixed-wing UAVs can be broadly categorized into two classes: sampling-based methods and optimal control-based methods.

Sampling-based methods [7], such as RRT [5] (rapidly-exploring random tree [9]) and its variants (e.g., RRT* [14]), have become mainstream techniques for trajectory generation in static or slowly changing environments due to their powerful obstacle avoidance and path exploration capabilities. These methods discretize and search the configuration space via sampling, enabling rapid generation of feasible paths, which is particularly suitable for global coarse planning. However, such approaches often lack incorporation of a complete dynamic model of the UAV, frequently relying instead on point-mass or simplified kinematic assumptions. As a result, the generated trajectories may suffer from issues in dynamic feasibility and control smoothness. Moreover, since time optimality and dynamic coupling effects are not fully considered, the resulting trajectories are often unsuitable for direct application in high-speed, highly-maneuverable fixed-wing UAV control.

Optimal control-based methods [12], on the other hand, stem from system dynamics and formulate the trajectory planning problem as a constrained optimization problem, solved via numerical optimization techniques such as direct methods (e.g., shooting and collocation methods [4]), indirect methods (e.g., Pontryagin's minimum principle [13]), and control parameterization. Among these, the control parameterization approach, which discretizes continuous control variables into a finite set of parameters, has proven effective in handling complex dynamic constraints and is commonly used in UAV trajectory optimization. Although these methods perform well in theoretical modeling, they still face two significant challenges.

First, conventional numerical optimization techniques struggle to efficiently handle free-time (i.e., time-varying domain) problems. Treating the terminal time as an optimization variable increases the nonlinearity and computational complexity of the problem, often leading to convergence difficulties or excessive computational load. Second, the various state constraints (e.g., full-flight altitude and speed limits) that must be satisfied during UAV operation are continuous-time inequality constraints. Traditional penalty function methods or relaxation techniques often require manual tuning of penalty weights, which can easily result in constraint violations or distortion of the objective function, severely compromising the reliability and accuracy of the solution.

Therefore, despite considerable progress in trajectory planning research, there remains a lack of systematic solutions that effectively balance time optimality, constraint satisfaction, and computational reliability for fixed-wing UAVs. In particular, current methods exhibit limitations in simultaneously addressing free-time domains and continuous-state constraints, characterized by low optimization efficiency and high risk of constraint violation, which hinders their practical application in engineering contexts.

1.3. **Main work and contributions.** To address the aforementioned challenges, this study proposes a time-optimal trajectory replanning method for fixed-wing UAVs based on an exact penalty function. The main contributions are summarized as follows:

1. A trajectory planning framework integrating control parameterization, time-scale transformation, and an exact penalty function is developed, converting the original problem into a parameter optimization problem over a fixed time domain, thereby significantly reducing computational complexity.
2. A dynamically adjusted exact penalty function mechanism is designed to incorporate continuous state constraints into the objective function, achieving strict satisfaction of constraints.
3. The idea of model predictive control (MPC) is incorporated to achieve a closed-loop replanning-tracking, enhancing the robustness of trajectory tracking.
4. Extensive experiments based on an F-16 aircraft model are conducted, quantitatively validating the engineering feasibility of the proposed method in terms of trajectory tracking accuracy, the trackability of control commands and constraint compliance.

2. Problem modeling.

2.1. **3-DOF kinematic modeling of fixed-wing UAVs.** A spatial point on the ground is chosen as the origin O_g of the ground coordinate system ($O_g x_g y_g z_g$). The $O_g x_g$-axis is aligned with a reference direction in the horizontal plane, while the $O_g z_g$-axis is perpendicular to the horizontal plane and points upward. The $O_g z_g$-axis is perpendicular to the horizontal plane and points upward; the $O_g y_g$-axis is then determined according to the right-hand rule [2]. The origin O_k of the flight path coordinate system ($O_k x_k y_k z_k$) is set at the UAV's center of mass. The $O_k x_k$-axis points in the direction of the UAV's ground velocity vector; the $O_k z_k$-axis lies in the vertical plane containing the $O_k x_k$-axis, is perpendicular to the $O_k x_k$-axis, and points upward; the $O_k y_k$-axis is perpendicular to the $O_k x_k z_k$-plane, with its direction determined by the right-hand rule [18].

External forces acting on the fixed-wing UAV include thrust T, aerodynamic drag D, lift L, and gravity $G = mg$ (where m is the UAV mass and g is the gravitational acceleration). Relevant Euler angles [16] are defined as follows: the flight path deflection angle χ, the flight path angle γ, and the velocity roll angle ϕ (the angle between the UAV's symmetry plane and the vertical plane containing the velocity v) [1]. A schematic is shown in Figure 1.

The normal overload n and the longitudinal overload n_x are expressed as

$$n = \frac{L}{mg},\ n_x = \frac{T - D}{mg} \tag{1}$$

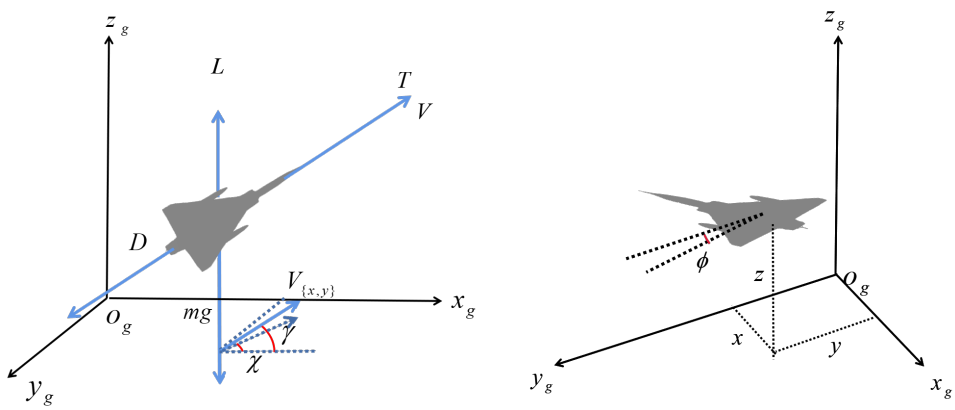

FIGURE 1. Schematic diagram of UAV forces and coordinate systems

By combining Newton's second law with the coordinate system transformation relationships, the 3-DOF dynamic model of the fixed-wing UAV can be derived as [6]:

$$\begin{cases} \dot{x} = v\cos\chi\cos\gamma \\ \dot{y} = v\sin\chi\cos\gamma \\ \dot{z} = v\sin\gamma \\ \dot{v} = gn_x - g\sin\gamma \\ \dot{\chi} = \dfrac{gn\sin\phi}{v\cos\gamma} \\ \dot{\gamma} = \dfrac{g\left(n\cos\phi - \cos\gamma\right)}{v} \end{cases} \tag{2}$$

The state vector is defined as

$$\boldsymbol{P}(t) = [x, y, z, v, \chi, \gamma, \phi, n]^\top \tag{3}$$

where $[x, y, z]$ are the positions in the ground coordinate system, v is the velocity, χ is the flight path deflection angle, γ is the flight path angle, ϕ is the roll angle, and n is the normal overload.

The control input vector is expressed as

$$\boldsymbol{u}(t) = \left[\dot{\phi}, \dot{n}, n_x\right]^\top \tag{4}$$

where $\dot{\phi}$ is the roll rate, $\dot{n}$ is the normal overload rate, and n_x is the longitudinal overload (directly affecting velocity changes).

$$\dot{\boldsymbol{P}}(t) = \begin{bmatrix} v(t)\cos\chi(t)\cos\gamma(t) \\ v(t)\sin\chi(t)\cos\gamma(t) \\ v(t)\sin\gamma(t) \\ gn_x(t) - g\sin\gamma(t) \\ \dfrac{gn(t)\sin\phi(t)}{v\cos\gamma(t)} \\ \dfrac{g\big(n(t)\cos\phi(t) - \cos\gamma(t)\big)}{v(t)} \\ \dot{\phi}(t) \\ \dot{n}(t) \end{bmatrix} \tag{5}$$

The initial condition is

$$\boldsymbol{P}(0) = [x_0, y_0, z_0, v_0, \chi_0, \gamma_0, \phi_0, n_0]^\top \tag{6}$$

where $[x_0, y_0, z_0]$ are the initial value of the positions in the ground coordinate system, v_0 is the initial value of the velocity, χ_0 is the initial value of the flight path deflection angle, γ_0 is the initial value of the flight path angle, ϕ_0 is the initial value of the roll angle, and n_0 is the initial value of the normal overload.

In practice, UAV maneuverability is limited by physical and structural constraints. For example, velocity, flight-path deflection angle, and flight-path angle cannot exceed

$$z_{\min} \le z \le z_{\max},\ v_{\min} \le v \le v_{\max},\ \gamma_{\min} \le \gamma \le \gamma_{\max}$$

$$\phi_{\min} \le \phi \le \phi_{\max},\ n_{\min} \le n \le n_{\max} \tag{7}$$

Here, $z_{\min}$ and $z_{\max}$ denote the minimum and maximum allowable altitude of the UAV, respectively; $v_{\min}$ and $v_{\max}$ represent the minimum and maximum feasible velocity constraints; $\gamma_{\min}$ and $\gamma_{\max}$ define the lower and upper bounds of the flight-path angle (the angle between the UAV's flight path and the horizontal plane); $\phi_{\min}$ and $\phi_{\max}$ specify the allowable range of the roll angle (the angle of rotation about the UAV's longitudinal axis); and $n_{\min}$ and $n_{\max}$ indicate the minimum and maximum load factor, which characterizes the aerodynamic forces (relative to the UAV's weight) that the vehicle can withstand during maneuvers.

Due to actuator limitations, the control variables must also satisfy

$$\dot{\phi}_{\min} \le \dot{\phi} \le \dot{\phi}_{\max},\ \dot{n}_{\min} \le \dot{n} \le \dot{n}_{\max},\ n_{x\min} \le n_x \le n_{x\max} \tag{8}$$

Here, $\dot{\phi}_{\min}$ and $\dot{\phi}_{\max}$ represent the minimum and maximum allowable roll rate (i.e., the rate of change of the roll angle ϕ, which describes how fast the UAV rotates about its longitudinal axis). $\dot{n}_{\min}$ and $\dot{n}_{\max}$ denote the minimum and maximum feasible rate of change of the load factor n (the load factor reflects the ratio of aerodynamic forces to the UAV's weight, and its rate of change constrains how rapidly this force ratio can vary during maneuvers). $n_{x\min}$ and $n_{x\max}$ specify the lower and upper bounds of the longitudinal load factor n_x (this is the load factor acting along the UAV's longitudinal direction, which is crucial for characterizing

longitudinal acceleration and structural loads during flight). The terminal state must satisfy the position constraints

$$x(T) = x_T,\ y(T) = y_T,\ z(T) = z_T \tag{9}$$

where x_T, y_T, and z_T denote the target position in the ground coordinate system, and T is the terminal time to be optimized.

The goal of trajectory planning is to minimize the total movement time, given the initial state, while considering the magnitudes of state and control variables and the target position constraints. The problem model (P1) is formulated as follows:

$$\begin{aligned}
&\text{(P1):} \quad \min_{\dot{n},\dot{\phi},n_x,T} T \\
&\text{s.t.} \quad \mathrm{C}_0\text{:} \quad \dot{\boldsymbol{P}}(t) = \begin{bmatrix} v(t)\cos\chi(t)\cos\gamma(t) \\ v(t)\sin\chi(t)\cos\gamma(t) \\ v(t)\sin\gamma(t) \\ gn_x(t) - g\sin\gamma(t) \\ \dfrac{gn(t)\sin\phi(t)}{v(t)\cos\gamma(t)} \\ \dfrac{g\big(n(t)\cos\phi(t) - \cos\gamma(t)\big)}{v(t)} \\ \dot{\phi}(t) \\ \dot{n}(t) \end{bmatrix} \\
&\qquad \mathrm{C}_1\text{:} \quad v_{\min} \le v(t) \le v_{\max} \\
&\qquad \mathrm{C}_2\text{:} \quad \gamma_{\min} \le \gamma(t) \le \gamma_{\max} \\
&\qquad \mathrm{C}_3\text{:} \quad \phi_{\min} \le \phi(t) \le \phi_{\max} \\
&\qquad \mathrm{C}_4\text{:} \quad n_{\min} \le n(t) \le n_{\max} \\
&\qquad \mathrm{C}_5\text{:} \quad z_{\min} \le z(t) \le z_{\max} \\
&\qquad \mathrm{C}_6\text{:} \quad x(T) = x_T \\
&\qquad \mathrm{C}_7\text{:} \quad y(T) = y_T \\
&\qquad \mathrm{C}_8\text{:} \quad z(T) = z_T \\
&\qquad \mathrm{C}_9\text{:} \quad \dot{\phi}_{\min} \le \dot{\phi}(t) \le \dot{\phi}_{\max} \\
&\qquad \mathrm{C}_{10}\text{:} \quad \dot{n}_{\min} \le \dot{n}(t) \le \dot{n}_{\max} \\
&\qquad \mathrm{C}_{11}\text{:} \quad n_{x\min} \le n_x(t) \le n_{x\max} \\
&\qquad \mathrm{C}_{12}\text{:} \quad \boldsymbol{P}(0) = [x_0, y_0, z_0, v_0, \chi_0, \gamma_0, \phi_0, n_0]^\top
\end{aligned}$$

where $t \in [0, T]$. In the above constraints, C_0 is the 3-DOF fixed-wing UAV dynamic equation constraint; C_1–C_5 are the state variable constraints; C_6–C_8 are the terminal state constraints; C_9–C_{11} are the control variable constraints; C_{12} is the initial state constraint. The time domain is variable, and constraints C_1–C_5 include

continuous time state constraints, which essentially generate an infinite number of constraints, posing significant challenges to problem solving.

3. **Optimal control algorithm based on exact penalty function.**

3.1. **Control parameterization.**

To address the optimization problem (P1), this study adopts a combination of time-scale transformation and control parameterization. The time-scale transformation maps the free-time domain $[0, T]$ into a fixed interval $[0, 1]$, where s denotes the scaled time (Figure 2):

$$\frac{\mathrm{d}t}{\mathrm{d}s} = T \tag{10}$$

For simplicity, Eq (2) is expressed as a function f, yielding the UAV dynamic model in state-space form

$$\dot{\boldsymbol{P}}(t) = f(\boldsymbol{P}(t), \boldsymbol{u}(t)) \tag{11}$$

Applying the time-scale transformation leads to

$$\dot{\boldsymbol{P}}(s) = \frac{\mathrm{d}\boldsymbol{P}}{\mathrm{d}s} = \frac{\mathrm{d}\boldsymbol{P}}{\mathrm{d}t} \cdot \frac{\mathrm{d}t}{\mathrm{d}s} = f(\boldsymbol{P}(s), \boldsymbol{u}(s)) \cdot T \tag{12}$$

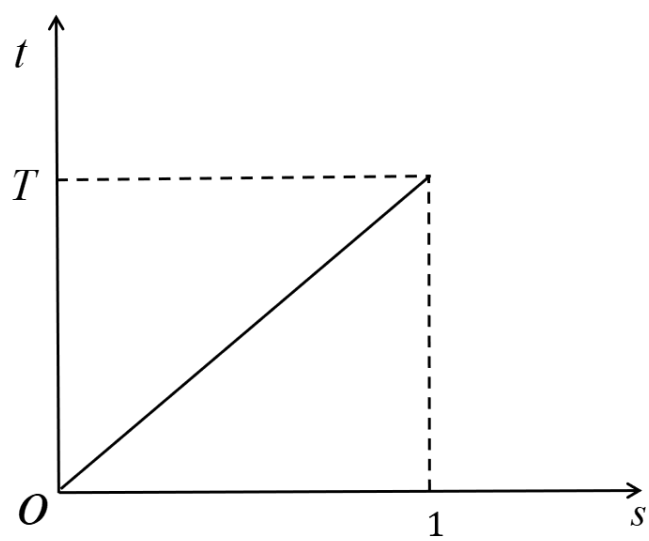

FIGURE 2. Time scale transformation

According to [8] and [10], for ease of optimization, the continuous control variables are divided into K equal intervals within $s \in [0, 1]$, generating $K + 1$ nodes $[0, s_1, s_2, \ldots, s_K]$, where $s_K = 1$. Piecewise constant parameterization of the control

variables is then introduced:

$$\begin{aligned}\dot{n}(s) &= \sum_{m=1}^{K} \boldsymbol{\sigma}_{1,m}\Gamma_{(s_{K-1},s_K)}(s)\\ \dot{\phi}(s) &= \sum_{m=1}^{K} \boldsymbol{\sigma}_{2,m}\Gamma_{(s_{K-1},s_K)}(s)\\ n_x(s) &= \sum_{m=1}^{K} \boldsymbol{\sigma}_{3,m}\Gamma_{(s_{K-1},s_K)}(s)\end{aligned} \tag{13}$$

In Eq (15), $\sigma_{i,m}(i = 1, 2, 3)$ are the value of constant parameters (to be optimized) of three control variables in the mth interval, and $\Gamma_{(s_{K-1},s_K)}(s)$ is the interval indicator function:

$$\Gamma_{(s_{K-1},s_K)}(s) = \begin{cases} 1, & s \in (s_{K-1}, s_K) \\ 0, & s \notin (s_{K-1}, s_K) \end{cases} \tag{14}$$

For example, the parametermization of n_x is shown in Figure 3.

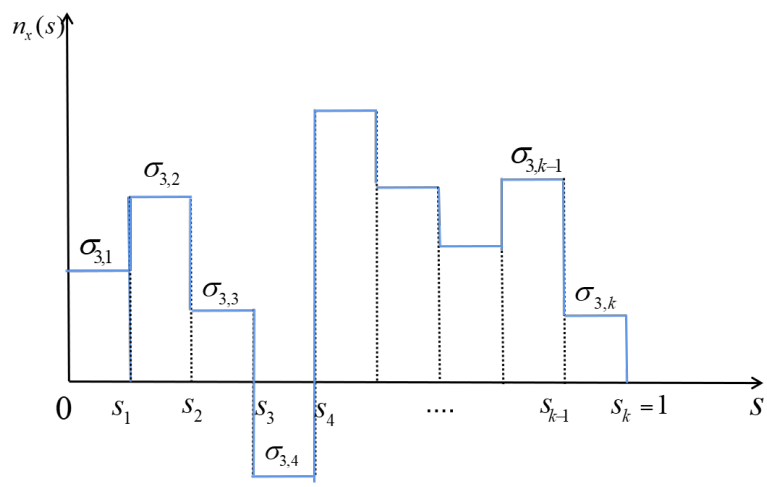

FIGURE 3. Control parameterization n_x

Through the above process, the infinite-dimensional optimal control problem (P1) is transformed into a finite-dimensional parameter optimization problem (P2).

$$
\begin{aligned}
&\text{(P2):} \quad \min_{\boldsymbol{\sigma},T} T \\
&\text{s.t.} \quad \mathrm{S}_0\text{:} \quad \dot{\boldsymbol{P}}(s) = \begin{bmatrix} v(s)\cos\chi(s)\cos\gamma(s)\cdot T \\ v(s)\sin\chi(s)\cos\gamma(s)\cdot T \\ v(s)\sin\gamma(s)\cdot T \\ \big(gn_x(s) - g\sin\gamma(s)\big)\cdot T \\ \dfrac{gn(s)\sin\phi(s)}{v(s)\cos\gamma(s)}\cdot T \\ \dfrac{g\big(n(s)\cos\phi(s) - \cos\gamma(s)\big)}{v(s)}\cdot T \\ \dot{\phi}(s)\cdot T \\ \dot{n}(s)\cdot T \end{bmatrix} \\
&\qquad \mathrm{S}_1\text{:} \quad v_{\min} \le v(s) \le v_{\max} \\
&\qquad \mathrm{S}_2\text{:} \quad \gamma_{\min} \le \gamma(s) \le \gamma_{\max} \\
&\qquad \mathrm{S}_3\text{:} \quad \phi_{\min} \le \phi(s) \le \phi_{\max} \\
&\qquad \mathrm{S}_4\text{:} \quad n_{\min} \le n(s) \le n_{\max} \\
&\qquad \mathrm{S}_5\text{:} \quad z_{\min} \le z(s) \le z_{\max} \\
&\qquad \mathrm{S}_6\text{:} \quad x(1) = x_T \\
&\qquad \mathrm{S}_7\text{:} \quad y(1) = y_T \\
&\qquad \mathrm{S}_8\text{:} \quad z(1) = z_T \\
&\qquad \mathrm{S}_9\text{:} \quad \dot{\phi}_{\min} \le \dot{\phi}(s) \le \dot{\phi}_{\max} \\
&\qquad \mathrm{S}_{10}\text{:} \quad \dot{n}_{\min} \le \dot{n}(s) \le \dot{n}_{\max} \\
&\qquad \mathrm{S}_{11}\text{:} \quad n_{x\min} \le n_x(s) \le n_{x\max} \\
&\qquad \mathrm{S}_{12}\text{:} \quad \boldsymbol{P}(0) = [x_0, y_0, z_0, v_0, \chi_0, \gamma_0, \phi_0, n_0]^\top
\end{aligned}
$$

where $s \in [0,1]$, and the optimization variables are $\boldsymbol{\sigma} = \left[\boldsymbol{\sigma_1}^\top, \boldsymbol{\sigma_2}^\top, \boldsymbol{\sigma_3}^\top\right]$ and T. Here, for each $i = 1,2,3$, $\boldsymbol{\sigma}_i = [\sigma_{i,1}, \sigma_{i,2}, \ldots, \sigma_{i,K}]$.

3.2. **Constraint handling based on exact penalty function.**

3.2.1. *Construction of the objective function.* Problem (P2) still involves terminal equality constraints (S_6–S_8) and continuous state inequality constraints (S_1–S_5), which make the optimization of P2 difficult. To address this, the exact penalty function method [20] is adopted, transforming (P2) into a problem with only control and initial constraints with finite dimensions. The objective function is redefined as

$$
\begin{aligned}
\min_{\boldsymbol{\sigma},T,\epsilon} J = \theta(\boldsymbol{P}(1), T, \epsilon) + \int_0^1 H(\boldsymbol{P}(s), \boldsymbol{\sigma}, T, \epsilon) - \boldsymbol{\lambda}^\top(s) f(s, \boldsymbol{P}(s), \boldsymbol{\sigma}) \cdot T \mathrm{d}s \\
\text{s.t. } S_9, S_{10}, S_{11}, S_{12}, \\
S_{13} : \epsilon > 0
\end{aligned} \tag{15}
$$

In Eq (15), $\theta(\cdot)$ handles the terminal position penalty term:

$$\theta(\boldsymbol{P}(1), T, \epsilon) = T + \epsilon^{-Q_1}\left[(x(1)-x_T)^2 + (y(1)-y_T)^2 + (z(1)-z_T)^2\right] \tag{16}$$

The Hamiltonian function is defined as

$$H(\boldsymbol{P}(s), \boldsymbol{\sigma}, \lambda(s), T, \epsilon) = \phi(\boldsymbol{P}(s), \boldsymbol{\sigma}, T, \epsilon) + \boldsymbol{\lambda}^\top(s) f(s, \boldsymbol{P}(s), \boldsymbol{\sigma}) \cdot T \tag{17}$$

where $\phi(\cdot)$ is the continuous state penalty term:

$$\begin{aligned}
&\phi(\boldsymbol{P}(s), \boldsymbol{u}(s), T, \epsilon) \\
&= \epsilon^{-Q_1}\Bigg[\max\left\{0, V(s) - V_{\max} - \epsilon^{Q_3}W\right\}^2 + \min\left\{0, V(s) - V_{\min} + \epsilon^{Q_3}W\right\}^2 \\
&\quad + \max\left\{0, z(s) - z_{\max} - \epsilon^{Q_3}W\right\}^2 + \min\left\{0, z(s) - z_{\min} + \epsilon^{Q_3}W\right\}^2 \\
&\quad + \max\left\{0, \chi(s) - \chi_{\max} - \epsilon^{Q_3}W\right\}^2 + \min\left\{0, \chi(s) - \chi_{\min} + \epsilon^{Q_3}W\right\}^2 \\
&\quad + \max\left\{0, \gamma(s) - \gamma_{\max} + \epsilon^{Q_3}W\right\}^2 + \min\left\{0, \gamma(s) - \gamma_{\min} + \epsilon^{Q_3}W\right\}^2 \\
&\quad + \max\left\{0, \phi(s) - \phi_{\max} - \epsilon^{Q_3}W\right\}^2 + \min\left\{0, \phi(s) - \phi_{\min} + \epsilon^{Q_3}W\right\}^2 \\
&\quad + \max\left\{0, n_z(s) - n_{z,\max} - \epsilon^{Q_3}W\right\}^2 + \min\left\{0, n_z(s) - n_{z,\min} + \epsilon^{Q_3}W\right\}^2\Bigg] \\
&\quad + Q_4\epsilon^{Q_2}
\end{aligned} \tag{18}$$

The costate dynamics are given by

$$\frac{\mathrm{d}\boldsymbol{\lambda}^\top(s)}{\mathrm{d}s} = -\frac{\partial H(\boldsymbol{P}(s), \boldsymbol{\sigma}, \boldsymbol{\lambda}(s), T, \epsilon)}{\partial \boldsymbol{P}(s)}, \ s \in [0, 1) \tag{19}$$

with terminal condition

$$\boldsymbol{\lambda}^\top(1) = \frac{\partial \theta(\boldsymbol{P}(1), T, \epsilon)}{\partial \boldsymbol{P}(1)} \tag{20}$$

In the above equations, ϵ is the penalty parameter [8]; $h(1)$ is the terminal state of the UAV; and Q_1, Q_2, Q_3, Q_4 (parameters of the exact penalty function), and ϵ are positive real parameters, and they should satisfy $2Q_3 - Q_1 - 1 > 0)$, $(Q_3 > Q_1$ and $-Q_1 - Q_2 + 2Q_3 > 0$; ϵ^{-Q_1} is the penalty weight; $Q_4\epsilon^{Q_2}$ is the penalty factor; and $\epsilon^{Q_3}W$ is the relaxation factor (where W is the weight of the relaxation factor) [11].

In the initial stage of optimization, due to the presence of numerous constraint violations, ϵ can be increased to enhance the penalty factor, thereby reducing the penalty weight and increasing the relaxation factor to relax the constraint penalty. As optimization progresses, the number of constraint violations gradually decreases, and ϵ can be reduced to decrease the penalty factor, thereby increasing the penalty weight and reducing the relaxation factor, ultimately further reducing constraint violations. When ϵ approaches 0, the penalty weight tends to infinity, the relaxation factor tends to 0, and the penalty factor tends to 0. At this point, the optimized objective function J can be regarded as tending to satisfy the constraints.

It should be noted that Q_4 theoretically needs to be infinitely large to make ϵ approach 0 during optimization. In practical calculations, ϵ can be considered sufficiently small when it is less than a certain threshold; thus, Q_4 can be set to

a large constant value. For the relaxation factor weight W, specific values can be adjusted according to the relaxation degree of different penalty terms; for the setting of other parameters, only the constraints required by the theory need to be satisfied.

The optimization problem has been transformed into a nonlinear programming (NLP) problem, and its gradient is as follows [19]:

$$\begin{aligned}
\frac{\partial J}{\partial \boldsymbol{\sigma}} &= \frac{\partial \theta(\boldsymbol{P}(1), T, \epsilon)}{\partial \boldsymbol{\sigma}} + \int_0^1 \frac{\partial H(\boldsymbol{P}(s), \boldsymbol{\sigma}, \boldsymbol{\lambda}(s), T, \epsilon)}{\partial T} \mathrm{d}s, \\
\frac{\partial J}{\partial T} &= \frac{\partial \theta(\boldsymbol{P}(1), T, \epsilon)}{\partial T} + \int_0^1 \frac{\partial H(\boldsymbol{P}(s), \boldsymbol{\sigma}, \boldsymbol{\lambda}(s), T, \epsilon)}{\partial T} \mathrm{d}s, \\
\frac{\partial J}{\partial \epsilon} &= \frac{\partial \theta(\boldsymbol{P}(1), T, \epsilon)}{\partial \epsilon} + \int_0^1 \frac{\partial H(\boldsymbol{P}(s), \boldsymbol{\sigma}, \boldsymbol{\lambda}(s), T, \epsilon)}{\partial T} \mathrm{d}s.
\end{aligned} \tag{21}$$

Using the gradients in Eq (21), this constrained nonlinear optimization problem can be efficiently solved by mature gradient-based NLP solvers.

4. Experimental validation.

4.1. Trajectory tracking mechanism based on model predictive control.

To mitigate tracking errors caused by model mismatch, this study incorporates model predictive control (MPC) [17] into the trajectory planning process. The proposed approach establishes a planning–tracking closed loop, as illustrated in Figure 4. At time $t = 0$, the trajectory planner obtains the UAV's real-time position and state information (including x, y, z, flight path angle, heading angle, velocity, normal overload, and roll angle), denoted as $\boldsymbol{P}(t)$, and plans a complete optimal trajectory from $\boldsymbol{P}(t)$ to the target (denoted as $\boldsymbol{P}_f$). Only the expected normal overload, roll angle, and velocity within the Δt period are sent to the controller, which controls the control surfaces to adjust the UAV's flight state and track the expected optimal trajectory. The UAV sends the real-time position and state information (denoted as $\boldsymbol{P}(t)$) after the Δt period to the parameter configurator. The parameter configurator exhaustively searches combinations of Q_1, Q_2, Q_3, and Q_4 (parameters of the exact penalty function), and outputs a set of parameter values that minimize the objective function (or make it less than a threshold) to the trajectory planner.

Then, the trajectory planner re-plans the trajectory using the selected Q_1, Q_2, Q_3, Q_4, and the UAV's current state $\boldsymbol{P}(t)$, and sends the expected normal overload, roll angle, and velocity within the next Δt period to the controller. The above steps are repeated continuously.

This cycle repeats until the remaining time between the current state $\boldsymbol{P}(t)$ and the target state $\boldsymbol{P}_f$ is shorter than Δt. At that point, a final trajectory segment is generated, and the mission concludes.

4.2. Experimental design and parameter configuration.

4.2.1. *Experimental platform and model selection.* The F-16 dynamic model [15] is adopted for validation. As a widely recognized benchmark in UAV dynamics and control, it accurately reflects key aerodynamic and dynamic characteristics, such as thrust–drag balance and lift–overload coupling, making it suitable for evaluating trajectory planning algorithms.

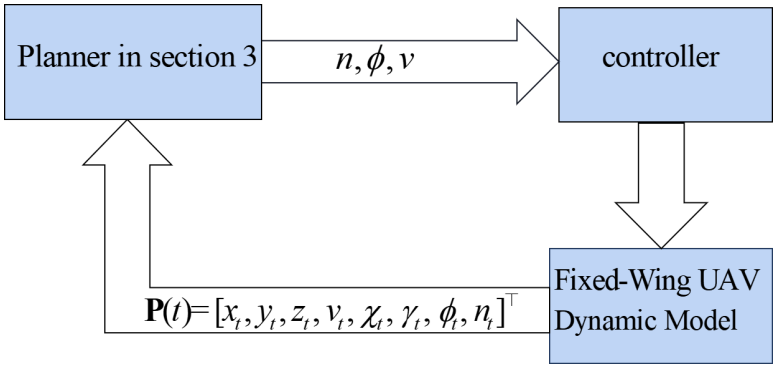

FIGURE 4. Schematic diagram of MPC-based trajectory replanning and tracking.

4.2.2. *Initial and terminal state settings.* To reflect medium- and high-altitude cruise and maneuver scenarios, the UAV's initial state is specified as $\boldsymbol{P}(0) = [x_0, y_0, z_0, v_0, \chi_0, \gamma_0, \phi_0, n_0]^\top = [1000, 1000, 7000, 180, \frac{\pi}{2}, 0, 0, 0.28]^\top$, where the initial flight path deflection angle $\chi_0 = \frac{\pi}{2}$ (pointing north in the ENU coordinate system), and the initial normal overload $n_0 = 0.28$ (corresponding to the cruise state).

The terminal position is $\boldsymbol{P}_f = [x_T, y_T, z_T]^\top = [1700, 4800, 7650]^\top$ (unit: m), with no additional attitude constraints on the terminal state (only position accuracy needs to be satisfied). The total flight time is 26.3 seconds.

4.2.3. *Constraint parameters and optimization configuration.* According to the actual maneuverability and structural limits of the F-16 aircraft, constraints on state and control variables are set as shown in Table 1. The rolling horizon Δt of MPC is set to 1 s. The exhaustive range of penalty parameters by the parameter configurator is $Q_1 \in [1.5, 3]$, $Q_2 \in [2.6, 4]$, $Q_3 \in [2.5, 4]$, and $Q_4 \in \{10^5, 10^6, 10^7, 10^8\}$, which satisfies the theoretical requirement of ϵ approaching 0. The optimal parameter combination is selected by minimizing the objective function J.

TABLE 1. Constraint parameters of state and control variables for fixed-wing UAV

Parameter	**Max Value**	**Min Value**	**Unit**
Altitude z	12000	2000	m
Velocity V	280	100	m/s
Flight path angle γ	$\frac{\pi}{6}$	$-\frac{\pi}{6}$	rad
Roll angle ϕ	$\frac{5\pi}{12}$	$-\frac{5\pi}{12}$	rad
Normal overload n	5	0.1	—
Roll rate $\dot{\phi}$	$\frac{\pi}{18}$	$-\frac{\pi}{18}$	rad/s
Normal overload rate $\dot{n}$	0.2	-0.2	—
Longitudinal overload n_x	1	-1	—

4.3. **Experimental results and analysis.**

4.3.1. *Trajectory tracking accuracy.* Regarding the spatial trajectory, as illustrated in Figure 5, the trajectory planned by the proposed method presents a smooth arc profile without inflection points, which is consistent with the dynamic characteristics of fixed-wing unmanned aerial vehicles; meanwhile, a total of 2630 sampling points (with an interval of 0.01) ($N = 2630$) were collected. For each sampling point, the Euclidean distance was employed to calculate the deviation between the actual position (x_i^r, y_i^r, z_i^r) and the planned position(x_i^p, y_i^p, z_i^p).

$$d_i = \sqrt{(x_i^r - x_i^p)^2 + (y_i^r - y_i^p)^2 + (z_i^r - z_i^p)^2} \tag{22}$$

where d_i denotes the spatial position error at the ith sampling point. Furthermore, the total length L_p of the planned trajectory was computed, defined as the cumulative sum of the distances between adjacent sampling points based on their planned positions.

$$L_p = \sum_{i=0}^{N} \sqrt{(x_{i+1}^p - x_i^p)^2 + (y_{i+1}^p - y_i^p)^2 + (z_{i+1}^p - z_i^p)^2} \tag{23}$$

and the average deviation $\bar{d}$ as

$$\bar{d} = \frac{1}{N} \sum_{i=0}^{N} d_i \tag{24}$$

The coincidence degree between actual and planned trajectories was then quantified as

$$C = 1 - \frac{\bar{d}}{L_p} \tag{25}$$

In the present simulation experiment, the obtained total length of the planned trajectory $L_p = 4046.72$ m, and the average positional deviation was 0.7237 m, yielding a coincidence degree close to 100%.

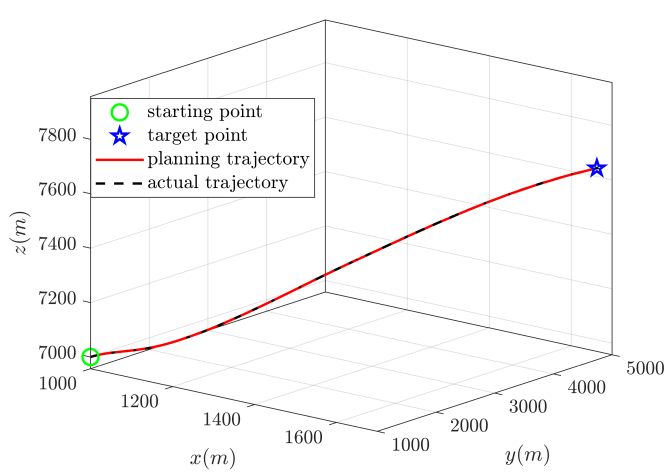

FIGURE 5. Spatial comparison between planned and actual trajectories

For state tracking performance, the proposed method exhibits high precision across key flight states: The maximum tracking error of the flight path angle γ (see Figure 6) is 0.0098 rad, occurring at 16.89 s; the flight path deflection angle χ (see Figure 7) has a maximum tracking error of 0.129 rad, observed at 18.92 s; the velocity v (see Figure 8) achieves a maximum tracking error of 7.93 m/s at 1.93 s without any overshoot behavior; the roll angle ϕ (see Figure 9) maintains a maximum tracking error of 0.1023 rad at 18.92 s, also free from overshoot; and the normal overload n_z (see Figure 10) shows a maximum tracking error of 0.2610 at 10.64 s, while its value strictly remains within the predefined range of $[0.1, 5]$ throughout the entire flight process.

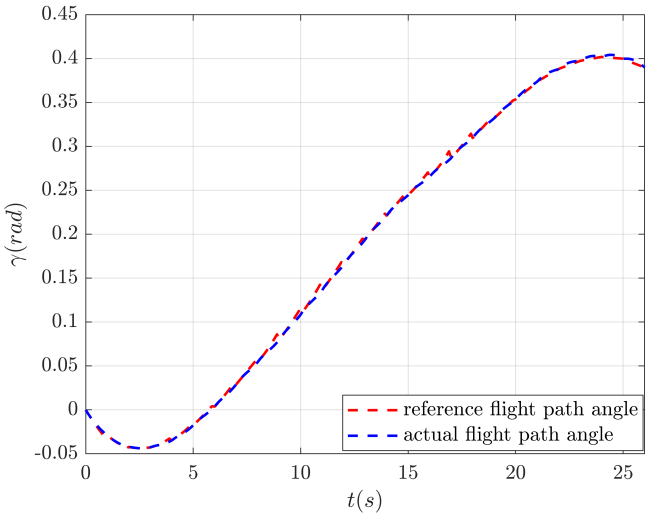

FIGURE 6. Comparison of reference and actual flight path angles

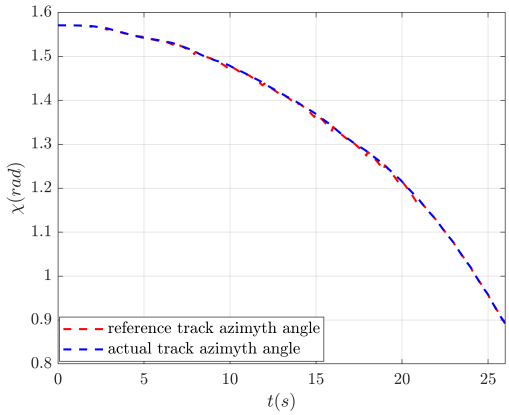

FIGURE 7. Comparison of reference and actual track azimyth comparison

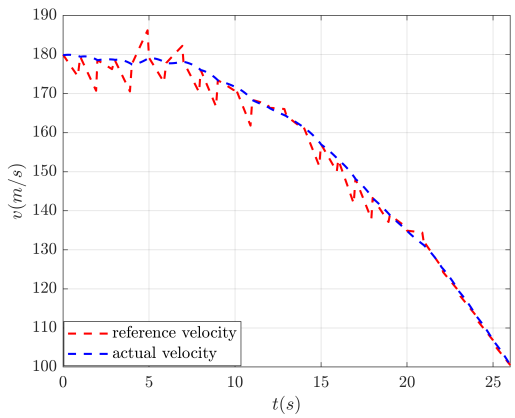

FIGURE 8. Comparison of reference and actual velocities

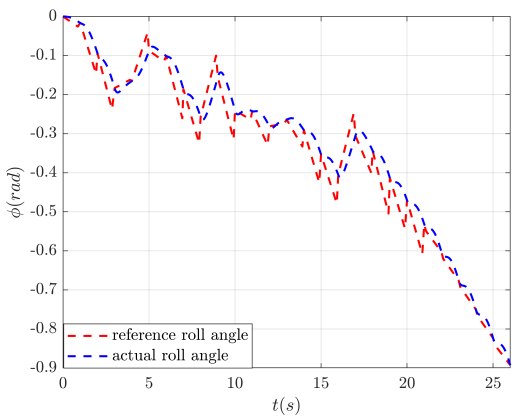

FIGURE 9. Comparison of reference and actual roll angles

As observed from the UAV trajectory tracking graphs, the trajectory planned by the exact penalty function and MPC perfectly coincides with the UAV's actual flight trajectory, indicating good trackability. The flight path deflection angle and flight path angle also coincide perfectly. The UAV performs a curved motion with a large turn, and when ϕ increases, the velocity should be reduced to avoid an excessively large turning radius; however, the overall requirement of minimal time is still satisfied.

4.3.2. *Constraint satisfaction.* Throughout the entire flight process, all control variables strictly adhere to their respective constraint limits. Specifically, the longitudinal overload n_x remains within the range $[-1, 1]$, with its maximum value reaching 1 only during the acceleration sprint phase (as illustrated in Figure 11). The normal overload rate $\dot{n}$ maintains stability within $[0.0438, 0.2]$, never exceeding the predefined boundary of $[-0.2, 0.2]$ (as illustrated in Figure 12). For the roll rate $\dot{\phi}$, its

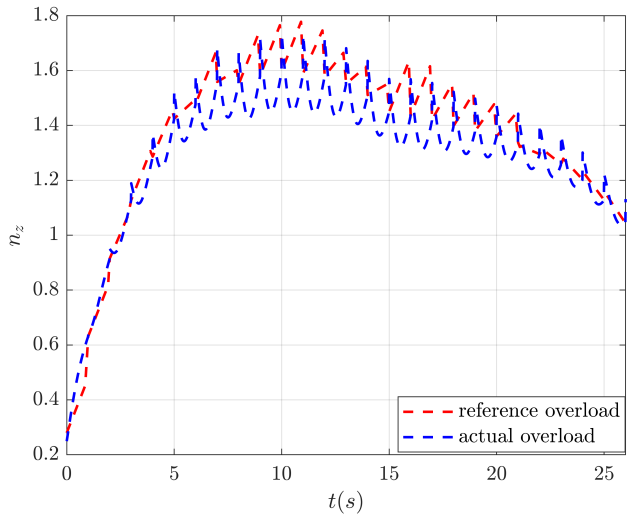

FIGURE 10. Comparison of reference and actual normal overloads

maximum magnitude is ±0.1745 rad/s, which is well within the allowable limit of ±0.1745 rad/s (as illustrated in Figure 13)

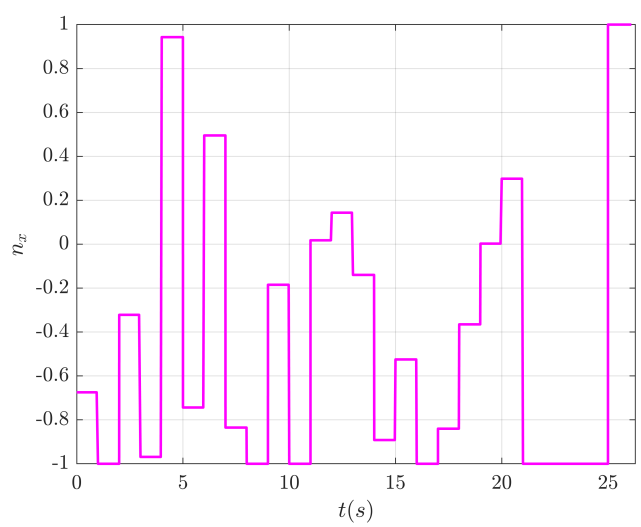

FIGURE 11. Input control signals-n_x

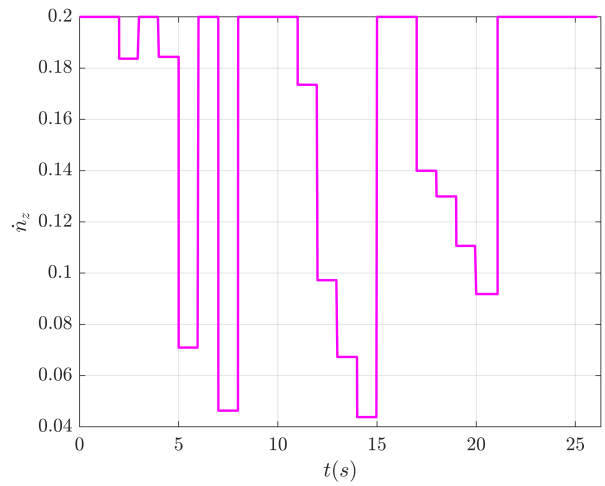

FIGURE 12. Input control signals-$\dot{n}_z$

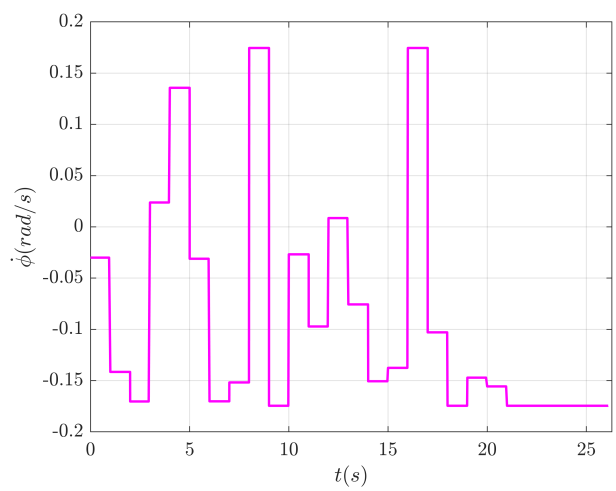

FIGURE 13. Input control signals- $\dot{\phi}$

These results confirm that the proposed exact penalty function combined with MPC produces a trajectory that is both dynamically feasible and strictly within physical constraints.

4.4. **Experimental conclusions.** Experiments based on the F-16 model demonstrate the following: The proposed method can effectively handle variable time domains and multiple continuous state constraints in time-optimal trajectory planning; the planned trajectory is smooth and conforms to the dynamic characteristics of fixed-wing UAVs; and physical constraints on state and control variables are strictly satisfied throughout the flight, with no constraint violations. Furthermore,

the control commands are smooth, indicating engineering implementability and traceability.

5. **Conclusion.** This study addresses the critical challenge of time-optimal trajectory planning for fixed-wing UAVs under complex dynamic and physical constraints. The core of our method lies in transforming the continuous control problem of a 3-DOF dynamic model into a finite-dimensional parameter optimization problem, with time optimality as the direct objective. A significant advantage of this formulation is the effective incorporation of state and control constraints into the objective function via a dynamic penalty parameter in exact penalty function, ensuring strict adherence to the UAV's physical limits. To bridge the gap between the simplified planning model and complete the 6-DOF model, the trajectory is tracked and replanned within an MPC framework. Comprehensive experimental validation based on the F-16 aircraft model conclusively demonstrates the feasibility and effectiveness of our proposed method.

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